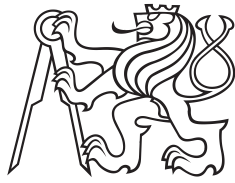


Master Thesis



Czech
Technical
University
in Prague

F3

Faculty of Electrical Engineering

Numerical Optimal Control for Minimum Lap Time

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Supervisor: doc. Ing. Zdeněk Hurák, Ph.D.
May 2026

Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 22. May 2026

Abstract

Thesis researches and implements ways of solving minimum maneuvering time problem (lap time simulation) for road vehicles. Thesis divides this problem into smaller subproblems, for which there are already mature software methods available. Subproblems are: racer car modelling, transcription of optimal control problem into NLP, solving of NLP, solving sets of linear equations and sensitivity analysis. Thesis implements these methods and benchmarks their performance. x

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Keywords: word, key

Supervisor: doc. Ing. Zdeněk Hurák, Ph.D.

Abstrakt

Diplomova práca skúma metódy riešenia minimum maneuvering time problemu pre cestne vozidla a implementuje ich. Práca delí tento problém na menšie podproblémy: vytvorenie modelu vozidla, zápis problému optimálneho riadenia, riešenie nelineárneho programu, riešenie systému lineárnych rovníc a citlivosťná analýza. Pre tieto podproblémy existuje mnoho rôznych možností riešenia, práca tieto možnosti vysvetľuje a porovnáva. V práci sú implementované metódy riešenia podproblémov ktoré som považoval za najužitočnejšie. Práca obsahuje aj benchmark vybraných možností riešenia.

Klíčová slova: NLP, laptime simulation, ipopt, skopirovat z bakalarky

Překlad názvu: Metody numerického optimálního řízení pro minimalizaci času na kolo — Cesta do tajů kdovíčeho

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Chapter 1

Introduction

1.1 Motivation

Topic of this thesis is motivated, by my work in Formula Student team eForce Prague Formula, where I worked on real-time vehicle dynamics control software. There I got intricately exposed to problems faced by racing teams, such as should we increase the aerodynamic downforce at the cost of higher drag? Should we put accumulator in the middle of vehicle, to decrease moment of inertia? Should the aerodynamics be designed for straight line drive, or for steady state turning? How to use limited energy from accumulator more effectively, how faster can we go if we made better power limiting algorithm? How large the accumulator should be? Which torque allocation algorithm should be used? How to design the suspension kinematics to maximize the tire utilization? Is rear wheel steering worth the increased weight? All of these problems can be analyzed by using purpose built tool, good engineering practices or using pen and paper with heavily simplified model. Formula student season is 10 months for development and construction of the vehicle, 2 months for competition. That puts enormous pressure on deciding on vehicle concept fast. In my opinion what formula student teams need is a tool to quickly evaluate different car concepts. As my bachelor thesis I created a prototype of such tool, goals of this thesis come from computational and user experience from my previous tool.

1.2 Goals

- Primary goal of this thesis is to investigate available methods for numerical optimal control with respect to their suitability for planning an optimal trajectory of a vehicle aiming to achieve the minimum lap time on a racing circuit. To do this for various vehicle concepts, thesis has to present a vehicle modelling framework.
- Benchmark performance of available numerical methods
- Thesis should present a tool for convenient vehicle modelling, and vehicle concept evaluation.

- Support vehicle parameter optimization, using any vehicle parameter as control input and sensitivity analysis of vehicle parameters.
- Goal is not precise vehicle modeling and validation, goal is to create a framework to perform vehicle concept analyses easily.

1.3 Structure of thesis

First part of thesis with vehicle modelling in the context of optimal control. A universal model structure inspired by [13] is presented. Then the theory of optimal control is presented.

1. ODE models of cars
2. Solving methods: time marching forward, backward pass, NLP,
3. Collocation (p, h, hp refinement), integral + differential form (Gauss-Radau, Legendre, Lobatto)
4. Implementation overview, Algorithmic differentiation, JUMP/Examples/InfiniteOpt + Julia, Python/MATLAB, CasADi, interior point solvers
5. Actual implementation and results
6. Conclusion

1.4 Literature overview/ survey of laptime simulators

Early attempts at lap-time computation started in 1950's. In 1970s computers became commonly used to solve MLTS using Quasi Steady State methods (QSS), these methods are what most race engineers think of when laptime simulation is mentioned. Here a car is modeled as point mass moving along predefined trajectory with lateral and longitudinal accelerations limited by steady state performance envelope[38], transient dynamics such as yaw acceleration are neglected. Formulations of MLTS as OCP appeared in late 1980s [23]. Article which summarizes history and state of the art methods of computing MLTS is [32]. Article [36] is widely regarded as the reference work, many parts of this thesis are taken from it. This article defines the track model, change of independent variable from time to distance traveled, and a twin-track Formula 1 model, with some parameters as decision variables. It is a very good description of most modern MLTS solving methods using NLP. Methods of computing minimum lap time is still active topic in race engineering, as for determining better setup of a race car for a given track, on a given day yields significant advantage. With more precise model the accuracy of setup improves, but also the computational time increases. In [2] authors deal with computational efficiency of computing minimum lap-time,

by using sequential convex programming. SEM SKOPIROVAT VECI Z BAKALRKY ZO STATE OF THE ART OVERVIEW

A very popular laptime simulator is OptimumLap from OptimumG [3], It is a free commercial tool advertised to achieve 10% accuracy with real vehicles. It uses Quasi Steady State methods (QSS) This is the standard tool for laptime simulations. Open source implementation of the same method [33], My implementation of this method in chapter ?? . QSS methods are very fast, but the models for which they can be used is equally very simple.

Popular open source C++ implementation of laptime simulator using nonlinear programming and automatic differentiaoton[31]. It uses IPOPT and CppAD[6]. Vehicle model is 6DOF twintrack. Other implementations are work from austria or germany[24] class project from Georgia institute of technology [1]



Part I

Theory

Chapter 2

Vehicle and Track modelling framework

In assignment of this is

A key aspect will be the selection of a sufficiently accurate yet computationally tractable model of the vehicle

There is not one model of vehicle that would fit all requirements of race engineers. For each problem, a different complexity of the vehicle model is sufficient, e.g. for analysis of power limit algorithm, the lateral dynamics of the vehicle can be heavily simplified and simple transcription and solving methods yield sufficient accuracy within a fraction of a second, this is described in ???. Whereas analysis of torque allocation requires the full complex model and more computationally expensive transcription and solving methods. Therefore, a vehicle modeling framework is needed to satisfy the needs of different types of analyses.

2.1 Model like in MSD

The structure of the vehicle model has to allow extending/changing components, so that different vehicle concepts can be simulated, with minimal coding. Book in [13] describes modelling framework that allows to create car components as modules with flow and effort interface. This common interface then heavily simplifies process of creating a car model, as modules can be reused and chained together easily. ADD DESCRIPTION OF THE BROWN GRAPH THEORY HERE. velocity goes one direction and forces go the other direction, must give better description.

Vehicle components are chassis, wheel pivot, tire, aerodynamics, suspension, motor

2.2 Tire - road interface modelling

Job of the tire model in this thesis is to calculate forces acting on tire from the relative velocity and angle between tire and road. Input of general tire model is 3 speeds and 3 angles, output is 3 forces.

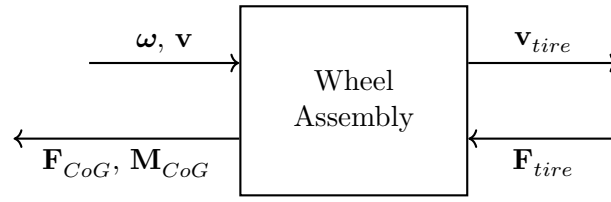


Figure 2.1: Wheel Pivot component: velocity inputs (top-left) transform to tire frame (top-right), tire forces (bottom-right) transform to forces and moments on CoG (bottom-left).

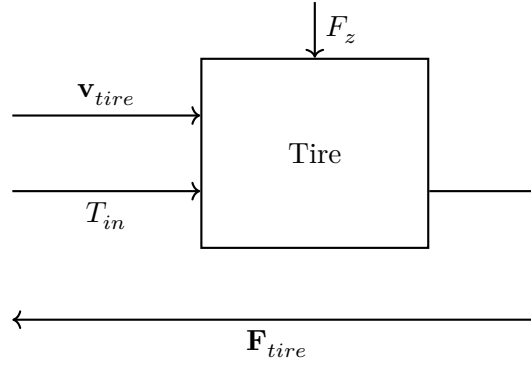


Figure 2.2: Tire component: tire-frame velocity $\mathbf{v}_{tire}$ and drive torque T_{in} on the left, normal force F_z from suspension on top, output is the tire force vector $\mathbf{F}_{tire}$ consumed by the wheel pivot.

Most forces that govern vehicle movement are transferred by the tires. The tire is considered to be the most important system of a vehicle and it is notoriously difficult to model [34]. Thus tires need to be modeled with great caution in mind. Not only are there many parameters, such as tire wear, tire temperature, road roughness, road wetness, tire pressure, etc., that affect the resultant forces, they are also constantly changing during the lap. Therefore they are difficult to measure and model precisely [14].

2.2.1 Linear tire

Simplest tire model. Tire is defined by friction ellipse and an angle at which the slip angle is highest

2.2.2 Pacejka

Pacejka tire uses

2.2.3 Extended Pacejka

2.3 Chassis model

point mass with definition of wheel pivot points.

$$\mathbf{M}_{CoG,i} = \mathbf{r}_i \times \mathbf{F}_i, \quad \mathbf{F}_i = \begin{bmatrix} F_{x_i} \\ F_{y_i} \\ 0 \end{bmatrix}. \quad (2.1)$$

$$\mathbf{F}_{CoG} = \sum_i \mathbf{F}_i + \begin{bmatrix} F_{\text{drag}} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{M}_{CoG} = \sum_i \mathbf{M}_{CoG,i}. \quad (2.2)$$

$$\dot{\mathbf{v}} = \frac{\mathbf{F}_{CoG}}{m} - \boldsymbol{\omega} \times \mathbf{v}, \quad \dot{\boldsymbol{\omega}} = \frac{\mathbf{M}_{CoG}}{I}. \quad (2.3)$$

$$\mathbf{x} = \begin{bmatrix} v_x \\ v_y \\ \psi \\ \omega_z \end{bmatrix}, \quad \dot{\mathbf{x}} = \begin{bmatrix} \dot{v}_x \\ \dot{v}_y \\ \dot{\omega}_z \end{bmatrix}. \quad (2.4)$$

2.4 Wheel Pivot model

Wheel pivot point is modelled as transformer. It defines the transformation of CoG velocities into tire frame and then transforms tire forces and torque into CoG frame. Angular velocity and position vectors are formed from yaw rate $\dot{\psi}$ and the longitudinal distance l_i of the axle from the center of gravity:

$$\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}, \quad \mathbf{r}_i = \begin{bmatrix} l_i \\ b/2 \\ 0 \end{bmatrix}, \quad (2.5)$$

where b is the track width. Velocity at each tire pivot point is:

$$\mathbf{v}_{p_i} = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}_i. \quad (2.6)$$

Velocities are then rotated from the pivot frame into the tire frame using the steering angle δ_i :

$$R_t = \begin{bmatrix} \cos \delta_i & -\sin \delta_i & 0 \\ \sin \delta_i & \cos \delta_i & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{v}_{t_i} = R_t \mathbf{v}_{p_i}. \quad (2.7)$$

Tire forces F_{tx} and F_{ty} are computed from $\mathbf{v}_{t_i}$ by the tire model. Forces are then rotated back to the pivot frame:

$$\begin{bmatrix} F_{x_i} \\ F_{y_i} \end{bmatrix} = \begin{bmatrix} \cos \delta_i & \sin \delta_i \\ -\sin \delta_i & \cos \delta_i \end{bmatrix} \begin{bmatrix} F_{tx} \\ F_{ty} \end{bmatrix}. \quad (2.8)$$

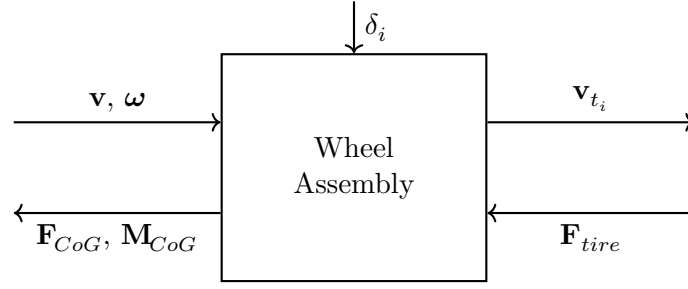


Figure 2.3: Wheel Pivot inputs and outputs. Steering angle δ_i is a control input. CoG velocity $\mathbf{v}$ and angular velocity $\boldsymbol{\omega}$ are mapped to tire-frame velocity $\mathbf{v}_{t_i}$, which is consumed by the tire model. Tire force vector $\mathbf{F}_{tire}$ is rotated and translated into forces $\mathbf{F}_{CoG}$ and moment $\mathbf{M}_{CoG}$ acting on the chassis.

2.5 Aerodynamics model

Aerodynamic model is modelled as nonlinear resistance element, mapping velocity into forces. Drag and downforce scale with the square of longitudinal velocity:

$$F_{\text{drag}} = \frac{1}{2} \rho C_D v_x^2, \quad F_{\text{lift}} = \frac{1}{2} \rho C_L v_x^2. \quad (2.9)$$

The downforce is split between front and rear axles by the center of pressure ratio $\text{CoP} \in [0, 1]$, where $\text{CoP} = 0.5$ corresponds to a balanced distribution. Air density ρ is taken as 1.225 kg/m^3 at sea level. Coefficients C_L , C_D and CoP used per vehicle model are listed in Table 2.1.

Table 2.1: Parameters of the aerodynamic model.

Symbol	Unit	Description
C_L	—	Lift coefficient (negative = downforce)
C_D	—	Drag coefficient
CoP	—	Center of pressure (front axle share, 0–1)
ρ	kg/m^3	Air density

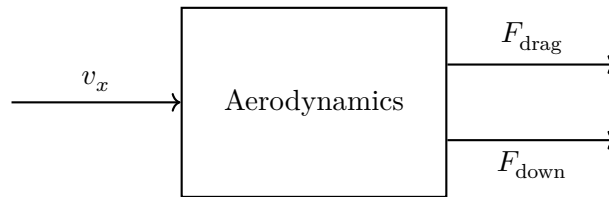


Figure 2.4: Aerodynamics component: longitudinal velocity v_x maps to drag force F_{drag} and downforce F_{down} .

This model is sufficient when vehicle is driving relatively straight, that means lateral acceleration is roughly zero. With high lateral acceleration this model is no longer valid, because air speeds are different on left and right part of car and also the car starts rolling- one side of vehicle is closer than the other, creating imbalance of forces.

2.6 Suspension model

Job of suspension in car is to maintain tire contact, distribute the load from lateral load transfer to better utilize tires and use kinematics to make use of camber thrust. Commonly suspension of car consists of spring, damper and the mounting with tie rods. In this thesis Suspension model had to be significantly simplified. Modelling each wheel with a damper, spring and its own kinematics is beyond the scope of this thesis. However, the structure of allows it, i just didnt have enough time to implement it. this quarter car model would be very beneficial toward the goal of scalability, now without it the model scalability is limited as divigin force between more than two wheel on axis is staticky nedoručeny problem.

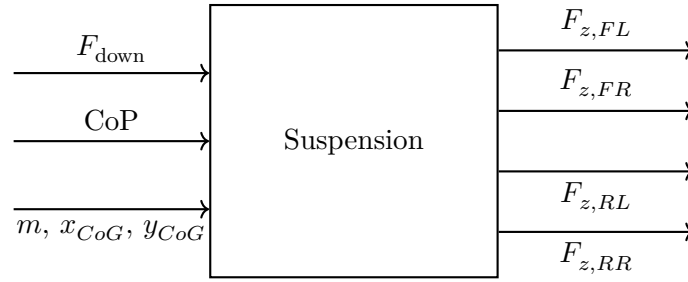


Figure 2.5: Simple suspension component: downforce F_{down} , center of pressure CoP and chassis geometry (mass m , CoG offsets $x_{\text{CoG}}, y_{\text{CoG}}$) map to the four wheel normal loads $F_{z,FL}, F_{z,FR}, F_{z,RL}, F_{z,RR}$.

2.7 Gearbox model

gearbox is the textbook example of transformer and resistive element.

$$T_{\text{out}} = \eta i T_{\text{in}}, \quad \omega_{\text{out}} = \frac{\omega_{\text{in}}}{i}. \quad (2.10)$$

Table 2.2: Parameters of the gearbox model.

Symbol	Unit	Description
i	—	Gear ratio (input to output speed)
η	—	Mechanical efficiency

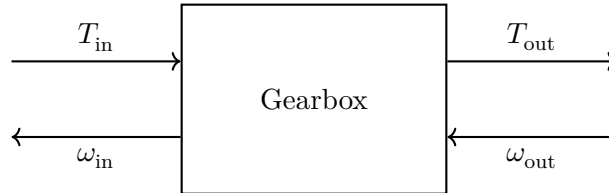


Figure 2.6: Gearbox component: motor-side torque T_{in} enters from the left, wheel-side angular velocity ω_{out} enters from the right; outputs are the geared-up torque T_{out} to the wheel and motor-side angular velocity ω_{in} to the motor.

2.8 Accumulator model

Accumulator is modelled as capacitor from brown, with additional constraints on maximum and minimum power

2.9 Motor model

$$P_{\text{mech}} = T \omega, \quad P_{\text{elec}} = \max\left(\frac{P_{\text{mech}}}{\eta}, P_{\text{mech}} \eta\right). \quad (2.11)$$

Table 2.3: Parameters of the motor model.

Symbol	Unit	Description
T	Nm	Shaft torque
η	—	Motor + inverter efficiency
P_{mech}	W	Mechanical shaft power
P_{elec}	W	Electrical port power (battery side)

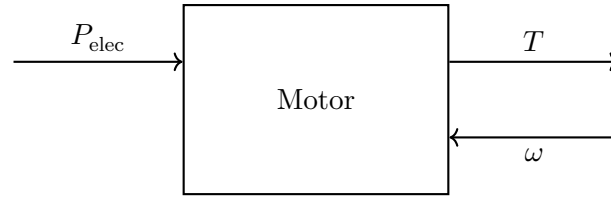


Figure 2.7: Motor component: electrical power P_{elec} enters from the battery side, shaft angular velocity ω is imposed by the drivetrain, output is shaft torque T .

2.10 Car models

These models must work when used in NLP transcription for solving boundary value problem, but they also must work for initial value problem, when the model is initialized or verified for transcription error. Some toolboxes can automatically create constraints from ODE models[45]. In my implementation I have and if statement that switches between IVP and BVP variant of models equations, major disadvantage of this approach is that the equations for both variants must be effectively the same and finding errors is difficult in this.

2.10.1 Generic car model

one of the goals is calability, therefore any car Parameter can be assigned as state, control, static parameter, decision variable or sensitivity analysis variable. This allows for simple RELAXATION? rozvolnenie? of DOF allowing to decide on one line of code which parameters should be controlled. This also allows for finding the optimal parameters of vehicle, such as brake bias,

aerodynamics balance, gear ratio. When parameter is chosen to be sensitivity analysis variable, impact of changing this variable is cheaply computed after the optimisation is finished, see Section 3.8.

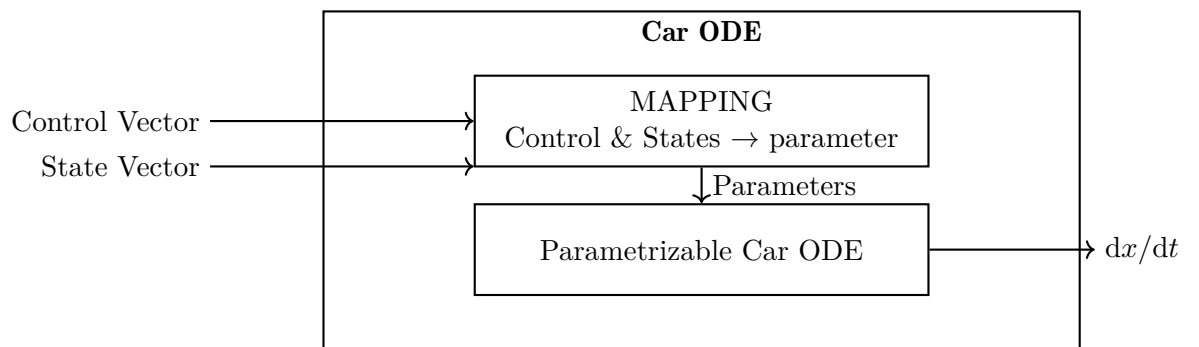


Figure 2.8: Generic car model structure. Control and state vectors are mapped to car parameters, which are then used by the parametrizable car ODE to compute derivatives.

2.10.2 Mass Point

2.10.3 Single track

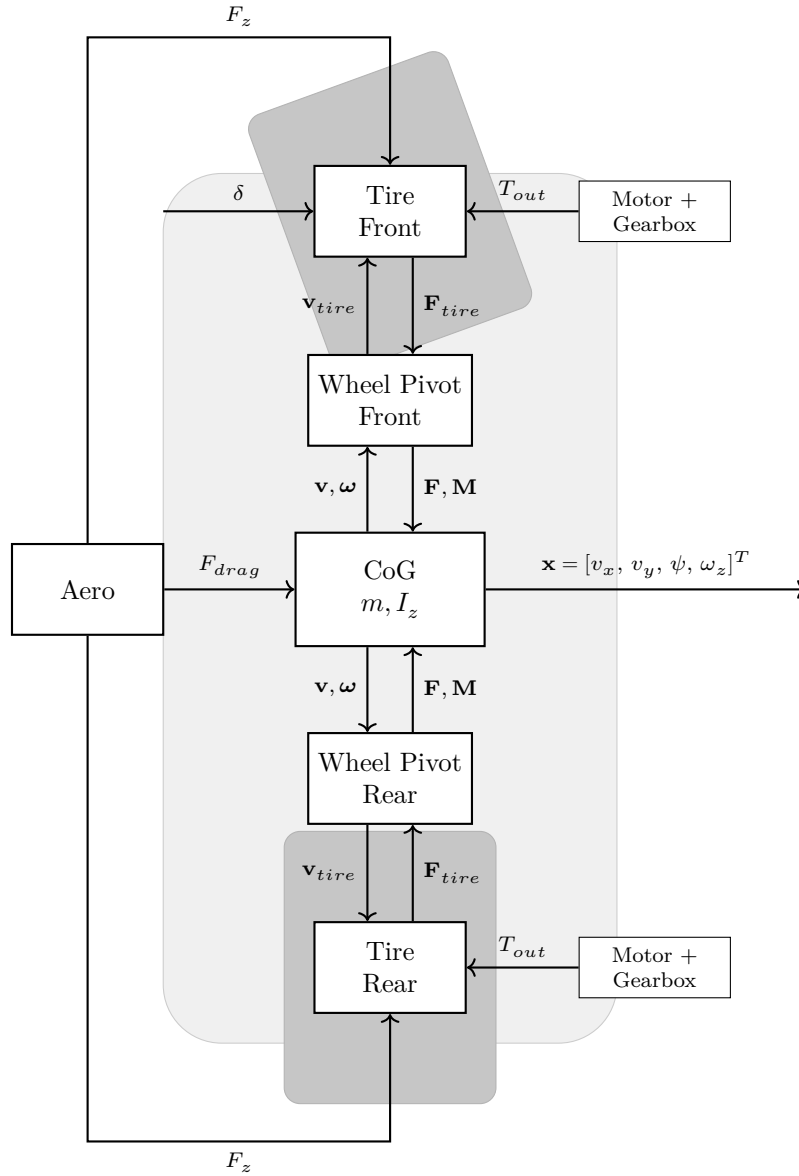


Figure 2.9: Single-track (bicycle) model computational graph. Velocity flows from CoG through wheel assemblies to tires, forces flow back. Motors provide torque through gearboxes, aerodynamics provides drag and normal forces.

2.10.4 Formula Student twin track

2.10.5 Formula Electric twin track

2.11 Track modeling

Track is sampled at discrete arc length nodes s_i and stored as a set of fields evaluated at each node:

- centerline coordinates $x(s)$, $y(s)$,
- heading $\theta(s)$,
- curvature $\kappa(s)$,
- left and right track widths $w_L(s)$, $w_R(s)$,
- air density $\rho(s)$,
- friction coefficient $\mu(s)$,
- inclination $\phi(s)$.

Each field is interpolated by a cubic spline in the arc length s , yielding C^2 -continuous fields along the entire trajectory, no nono no this did not work well, loinear interpolation was better. Obtaining suitable track for optimization from GPS is non-trivial, simple methods lead to noisy data, which then corrupts the optimization result. Therefore smoothing method described in [36] has been used.

2.12 Change of independent variable, time to path

When differential equations for vehicle movement are stated. For use in lap time simulation it is benefecial to describe the car states not in time but in distance traveled. This introduces connection with location on track and reduces number of state variables by one. This tranformation is described in[36].

$$\varepsilon = \psi - \theta(s) \quad (2.12)$$

$$S_f = \frac{dt}{ds} = \frac{1 - n C(s)}{v_x \cos \varepsilon - v_y \sin \varepsilon} \quad (2.13)$$

$$\frac{dn}{dt} = v_x \sin \varepsilon + v_y \cos \varepsilon \quad (2.14)$$

$$\frac{d}{ds} \begin{pmatrix} v_x \\ v_y \\ \psi \\ \omega \\ n \\ t \end{pmatrix} = S_f \begin{pmatrix} \dot{v}_x \\ \dot{v}_y \\ \dot{\psi} \\ \dot{\omega} \\ \dot{n} \\ 1 \end{pmatrix} \quad (2.15)$$

Chapter 3

The Optimal Control Problem

In previous chapter I have described how to build differential equations that govern movement of vehicle. Methods discussed in this section are doing essentially the same thing, solving controlled ODE along a time/path trajectory. But their approach and results are different.

3.1 Solving controlled ODE, initial value problems and boundary value problems

USE BOUNDARY AND INITIAL VALUE PROBLEM INSTEAD OF TIME MARCHING
The most common method for solving controlled ODE is time marching. Where the state computation is always for the states at time $t+1$, after solving at $t+1$, time is moved/marched forward and solution at $t+2$ is calculated and so on. This gives a solution to ODE. If the ODE can be controlled like the car, you also need a separate controller, which becomes part of the ODE. This approach gives a solution to the problem of how to control car, so that it completes a lap around a circuit. However, this solution is not optimal to time or any objective as the controls depend on the controller. Therefore, if the car parameters change and the controller doesn't, the resulting time can be exactly same, because controller cannot take advantage of lighter car. Sensitivity analysis and lap time can be obtained but it is skewed by the controller chosen. In Car maker the controller is driver and they have Driver adaptation. Using time marching is not best for sensitivity analysis and lap time simulation. Controller can be tuned after each change of model but, that is time intensive. Another approach to solving controlled ODE is using shooting methods. Now the algorithm doesn't compute the states at only next time point, but it computes all states at all time points at the same time. Using this method it is possible to get rid of the controller, and instead of control rule a feed forward control vector along the trajectory is computed. This chapter deals with transcribing the continuous problem into discrete steps

3.1.1 Initial value problem

3.1.2 Forward backward pass

DESCRIBE POPULAR LAPTIME SIMULATION METHOD THAT USES FORWARD AND BACKWARD PASS FOR A MODEL OF LOW COMPLEXITY

3.2 Transcription methods

3.2.1 RK methods

Most straightforward method to transcribe continuous optimal control problem into non linear program, is to use explicit Runge Kutta solving methods from initial value problem as equality constraints on the dynamics of vehicle. Using euler method, the constraints for the dynamics of vehicle:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h \cdot f(\mathbf{x}_k, \mathbf{u}_k) \quad (3.1)$$

This works, but it may be the worst possible transcription method, due to high error and instability of euler method [?]. Problems of euler method encountered in time marching simulation also translate to nonlinear programming. Implicit RK methods can be used too, In Initial value problems they are used less often, as they require iterative solving. Optimal control problem is Boundary value problem and not initial value problem, therefore:

...for a boundary value problem ...any scheme becomes effectively implicit. Thus, the distinction between explicit and implicit initial value schemes becomes less important in the BVP context.

[9, p. 69]

For boundary value problem it makes no sense to use explicit Runge kutta methods, as they have inferior stability compared to implicit runge kutta methods.[37, p. 4]

For Boundary value problems commonly implicit runge kutta method used is from LobattoIIIA family, but any other method can be used. Book [12] uses only these.

3.2.2 Collocation methods

The concept of collocation is old and universal in numerical analysis ...For ordinary differential equations it consists in searching for a polynomial of degree s whose derivative coincides ("co-locates") at s given points with the vectorfield of the differential equation ...[4, p. 224]

Job of collocation method is to find a polynomial that approximates the solution of BVP along independent variable(most commonly time). Collocation

methods are a subset of implicit runge kutta methods. The main distinction between runge kutta methods and collocation. Is that the result of collocation is always polynomial, that does not hold for general Runge Kutta method. Collocation methods are the core of this thesis, I have implemented Gauss radau, gauss, gauss lobatto. Collocation methods differ in points at which the derivative of polynomial coincides ("co-locates") with the vector field of differential equation. Commonly used methods in optimal control are radau, lobatto, legendre. They can be written in integral and differential form[18]. Collocation methods in this thesis are based on Framework described in[19]

Polynomial that approximates the solution of ODE is based on Legendre polynomials. Important note that in actual implementation this calculation of polynomial basis is impractical, therefore barycentric form from [10] has been used as implementation of these methods.

$$L_i(\tau) = \prod_{\substack{j=0 \\ j \neq i}}^N \frac{\tau - \tau_j}{\tau_i - \tau_j}, \quad i = 0, \dots, N \quad (3.2)$$

$$x_j^N(\tau) = \sum_{i=0}^N x_{ij} L_i(\tau) \quad (3.3)$$

the differentiation matrix is made also using barycentric form from [10, p.513] Many articles omit, that this barycentric form is crucial for fast code.

$$\dot{x}_j^N(\tau_k) = \sum_{i=0}^N x_{ij} \dot{L}_i(\tau_k) = \sum_{i=0}^N D_{ki} x_{ij}, \quad D_{ki} = \dot{L}_i(\tau_k) \quad (3.4)$$

then the output?? of differential equation can be approximated using the differentiation matrix.

$$\dot{x}_j^N(\tau_i) = (D\mathbf{X})_{ij} \quad (3.5)$$

This is then used as constraints and voila nonlinear optimal control problem is born. 🤖

$$\begin{aligned} \min \quad & \Phi(\mathbf{X}_{N+1}) \\ \text{s.t.} \quad & D\mathbf{X} = F(\mathbf{X}_{\text{LG}}, \mathbf{U}_{\text{LG}}) \\ & \mathbf{X}_{N+1} = \mathbf{X}_0 + \mathbf{w}^T F(\mathbf{X}_{\text{LG}}, \mathbf{U}_{\text{LG}}) \\ & \mathbf{X}_0 = \mathbf{x}_0 \end{aligned} \quad (3.6)$$

3.3 Gauss methods as pseudospectral collocation

I have implemented Gauss-Legendre, Gauss-Lobatto, Gauss-Radau as global collocation methods, that is one polynomial approximating the entire trajectory. I have implemented them according to [18]

■ 3.4 Gauss methods as h-method

■ 3.5 Diving problem into many lower degree polynomials

■ 3.6 Mesh refinement methods

■ 3.7 Mesh refinement

■ 3.7.1 collocation integral form

integral form is straight forward, same as other Runge-Kutta Methods

■ 3.7.2 collocation differential form

differential form requires to create differentiation matrix of the approximating polynomial, which contains the derivative of the approximating polynomial.

■ 3.7.3 p adaptive methods

p adaptive methods work by increasing the order of polynomial, until the error is below user defined threshold. Figure ?? shows that error along track is not distributed equally. Error is localised around areas with large jump in controls. These areas need finer sampling. P methods cannot focus these specific points directly, they focus entire trajectory.

The errors are concentrated around the discontinuity of controls, where the behaviour of the vehicle is more complex/ requiring high order or finer approximation. Figure shows this??

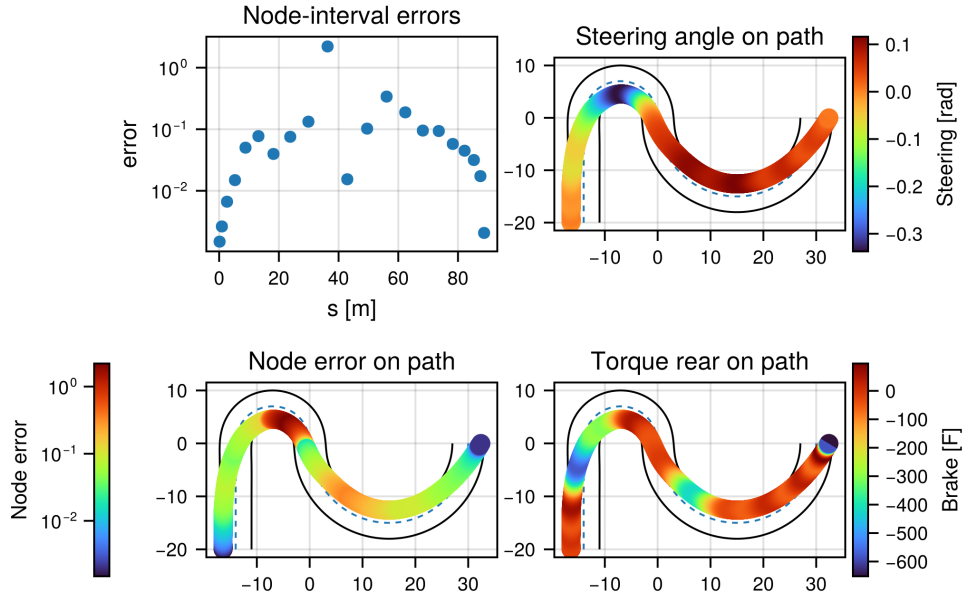


Figure 3.1: Node-interval ODE error for power-test case.

3.7.4 h adaptive methods

h methods are used when solution of optimal control problem is approximated by many intervals with low degree polynomial. These methods can decide to split segment into two smaller segments, they use different techniques to do this. Method described in [12] calculates error using DOPLNIT. Another method described in [26] uses techniques from essentially nonoscillatory (ENO) interpolation

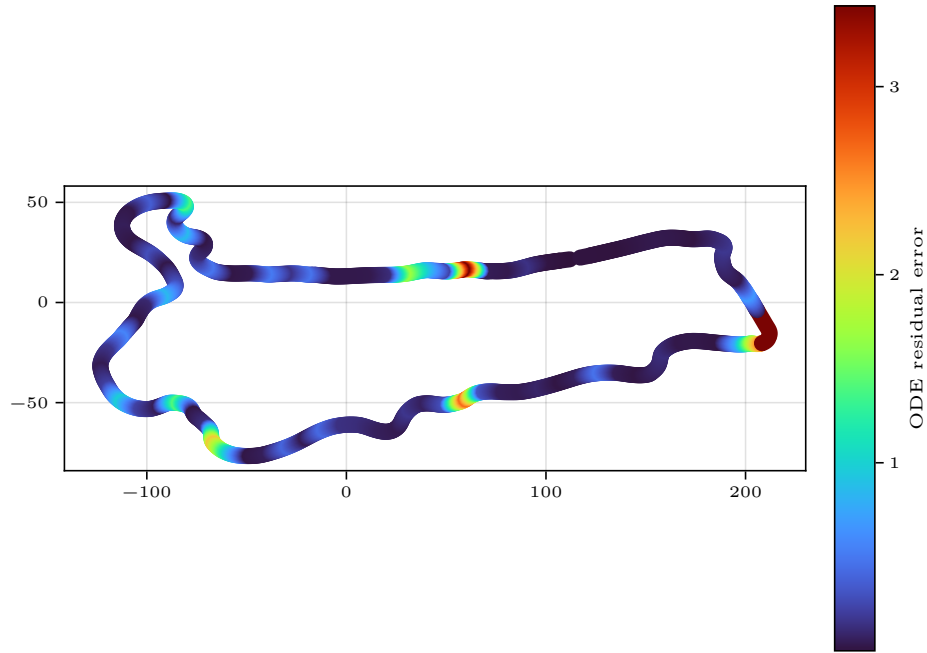


Figure 3.2: Node-interval ODE error for power-test case.

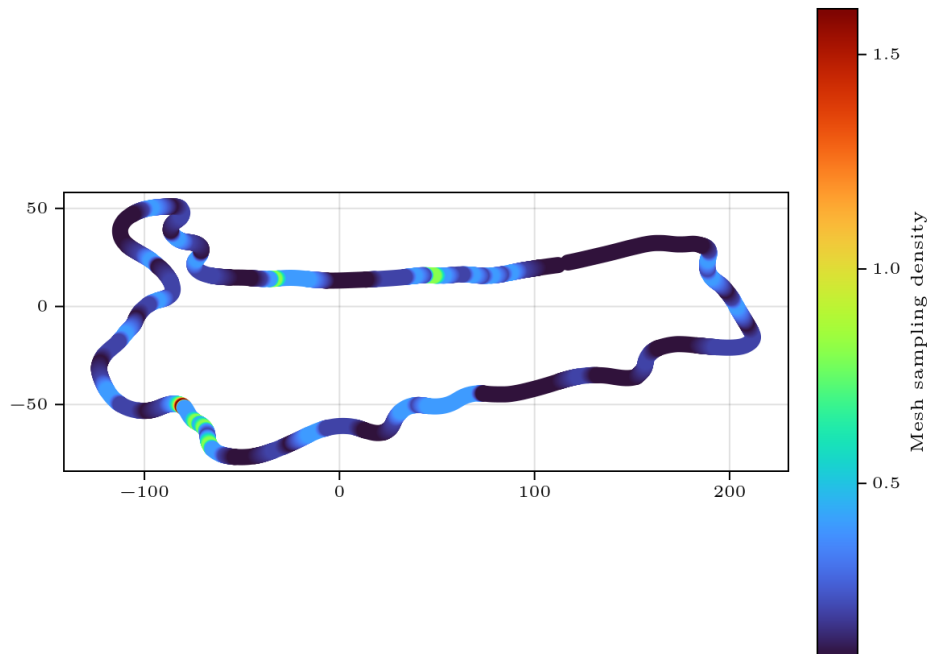


Figure 3.3: Node-interval ODE error for power-test case.

[22] to decide where to split and remove nodes.

■ 3.7.5 hp adaptive methods

hp adaptive methods are state of the art in mesh refinement. They are combination of increasing/decreasing mesh segments and increasing/decreasing order of approximating polynomial inside segments. One of the first hp-method and commonly cited is described in article [16] from University of Florida. They have produced numerous papers on hp- adaptive mesh refinement [7] [29]. Recent work on mesh refinement for functions with kinks [46]. They are the most complex for implementation and unfortunately I did not have the time to implement them.

■ 3.8 Adjoint methods

Adjoint methods allow for differentiating the solution of optimization problem. In this case it is function of time on states, controls and car parameters along the trajectory. This is very useful as it greatly simplifies sensitivity analysis. Traditional way of sensitivity analysis is to compute several runs with different vehicle parameters, but this allows to compute sensitivity of all parameters in one shot. I am using DiffOpt.jl [11], it is not exactly adjoint I don't really understand very well how it works

■ 3.8.1 quadrature rules

Chapter 4

NLP solvers, GPU exploitation???

Once the objective function and constraints are created, the problem is passed to NLP solver. Very popular methods are sequential quadratic solvers and interior point solvers.

4.1 Nonlinear programming

General NLP problems can be written as:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{h}(\mathbf{x}) = 0 \\ & \mathbf{g}(\mathbf{x}) \leq 0 \end{aligned} \tag{4.1}$$

4.1.1 KKT conditions

alpha and omega of nonlinear programming, lord praise the KKT conditions.

$$\begin{aligned} \nabla f(\mathbf{x}) + \sum_{i=1}^m \lambda_i \nabla h_i(\mathbf{x}) + \sum_{i=1}^p \mu_i \nabla g_i(\mathbf{x}) &= 0 \\ \mathbf{h}(\mathbf{x}) &= \mathbf{0} \\ \mathbf{g}(\mathbf{x}) &\leq \mathbf{0} \\ \mu_i &\geq 0, \quad i = 1, \dots, p \\ \mu_i g_i(\mathbf{x}) &= 0, \quad i = 1, \dots, p \end{aligned} \tag{4.2}$$

$$\begin{pmatrix} \nabla^2 \mathcal{L} & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} -\Delta x \\ \lambda \end{pmatrix} = \begin{pmatrix} \nabla f(x) \\ c(x) \end{pmatrix} \tag{4.3}$$

4.2 Interior point methods: IPOPT, MadNLP

Interior point solvers or barrier solvers work by simplifying the NLP into a barrier problem and solving a sequence of barrier problems, with decreasing barrier parameter. Solver used in this thesis is IPOPT [47]. Article in [43] describes framework that unifies algorithmic components of different solvers, allowing for custom new solvers.

4.2.1 Barrier problem formulation

There are multiple ways to transform NLP into barrier problem, this section will briefly describe how IPOPT does it. IPOPT article [47] Describes how to transform generic NLP into form 4.4 $h(\mathbf{x}) \geq 0$, interior point methods solve a barrier problem.

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \phi_\mu(\mathbf{x}) := f(\mathbf{x}) - \mu \sum_{i=1}^n \ln(x_i^{(i)}) \\ \text{s.t.} \quad & c(\mathbf{x}) = 0 \end{aligned} \quad (4.4)$$

Then barrier problem is further transformed into set of linear equations, that compute newton step.

$$\begin{bmatrix} W_k & A_k & -I \\ A_k^T & 0 & 0 \\ Z_k & 0 & X_k \end{bmatrix} \begin{bmatrix} d_x^k \\ d_\lambda^k \\ d_z^k \end{bmatrix} = - \begin{bmatrix} \nabla f(x_k) + A_k \lambda_k - z_k \\ c(x_k) \\ X_k Z_k e - \mu_j e \end{bmatrix} \quad (4.5)$$

There are numerous commercial and free algorithms to solve this. For smaller systems it is LU decomposition, but for larger sparse problems a dedicated solver is a must. IPOPT is most commonly used with MUMPS [8] solver which is free. There are several linear commercial solvers, Harwell Subroutine Library(HSL) supplies ma27 ma87 ma97 [25] pardiso solver [39], Harwell Subroutine Library solvers are available free for academic usage and pardiso solver is available free for academic but only single core. Latter solvers run on CPU, MadNLP solver is able to leverage GPU. Linear solver are capable of multithreading on CPU, which needs to be explicitly turned on. Tests were performed on Ryzen 9 9950X and MadNLP ran on Nvidia RTX 5070

4.3 Sequential quadratic programming

Sequential quadratic programming is less commonly used for offline trajectory optimisation, usage of IPOPT prevails.

4.4 Hessian calculation through Algorithmic differentiation

Major complication of many NLP solvers is that they require Hessian of the objective function in order to compute the step. computation of it can be time intensive. Most universal method is finite difference, that means evaluating the function around current point and estimating the hessian. other method is using symbolic differentiation, often the full symbolic expression is too large to be evaluated effectively due to chain and product rule. Algorithmic differentiation aims to solve problems of previous methods, by applying the chain rule to the elementary operations of a function. Comprehensive review of automatic differentiation is in article [15]

4.5 Scaling and matrix conditioning

Common pitfall of nonlinear programming is poor scaling, this happens when the solver encounters variables, that are orders of magnitude apart, for example motor force in thousands of newton and heading angle in radians. This happens, because at the core of nonlinear programming solver is a linear solver (MUMPS, MA97, PARDISO,...), that has to do matrix factorization every iteration, to solve KKT linear system. If this linear system is ill conditioned due to improper scaling, the factorization may be imprecise and search direction corrupted.

Another thing which led to issues is constraints qualification. Linear independence constraint qualification requires gradients of the active constraints to be linearly independent, then the jacobian of the constraints is rank deficient and linear solve may fail.

4.6 sparsity

Sparsity is good, sparsity means matrix Hessians and Jacobians with many zeros also sparsity in constraints, that means not having super very long expression with many terms in constraint. I guess it is because derivative is difficult then and can lead solver in wrong direction or take small steps. Therefore using more decision variables to define same problem can actually lead to faster computation [28]. Certain solvers can exploit the sparsity structure, such as FATROP [44]. In source [27] Joris Gillis a Casadi developer compares IPOPT to FATROP, with **FATROP being 33 times faster on a trajectory optimisation problem.**

Also [12] describes how sparsity structure helps with faster Hessian approximation for using finite difference??? secant method.

4.7 Initialization

NLP solvers require initial guess for all decision variables, to start solving. For this problem initial guess is obtained driving the vehicle on given track. PD controller is used to keep the car on the centerline and P controller is used to keep velocity at 5m/s. In general, it is possible to initialize all variables at zero, but with large scale nonlinear programming this approach is not used. For this problem initialization with zeros leads to singularity in equation 2.13 and is not possible at all. must be initialized otherwise very bad?

4.8 Global and local optimums

This optimization problem is generally non convex, therefore any local optimum does not guarantee global optimum. Commonly used NLP solvers, like

4. *NLP solvers, GPU exploitation???*

IPOPT guarantee only local optimums and not global ones. Good initialization is basis for finding a local optimum close to the global optimum. Source?????



Part II

Implementation of Optimal Control problem

Chapter 5

Frameworks for Optimal Control Problem Formulation

Process of solving optimal control problem has 3 parts. Problem definition, solving of nonlinear program, optional mesh refinement

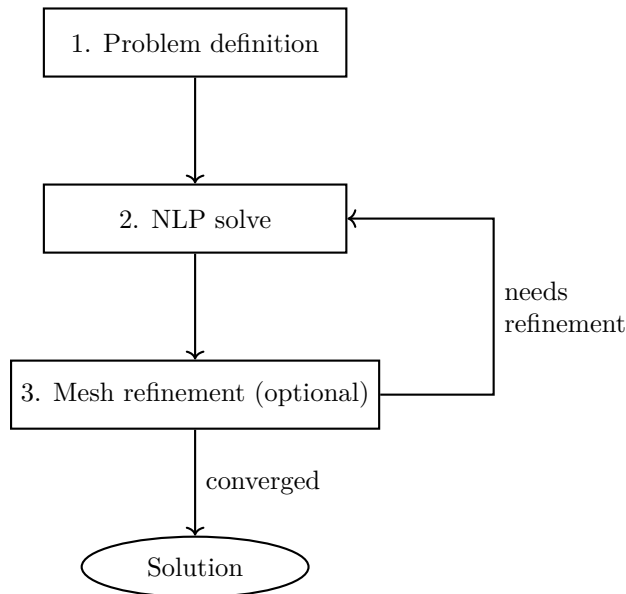


Figure 5.1: Optimal control problem solving pipeline.

The consensus in optimal control community about tools for problem formulation is that:

There is no shortage of optimal control software based on pseudospectral approaches. There are a number of other optimal control libraries that tackle similar kinds of problems, such as OTIS4, GPOPS-II, and CASADI.

[17][?] This chapter is about tool that define the OCP, and supply derivatives of the problem, they cannot solve the problem by themselves.

5.1 GPOPS-II

GPOPS-II [35] is perhaps the most advanced tool for optimal control, it was developed at University of Florida by Michael Patterson and Anil V. Rao. Other tools allow defining the problem, but GPOPS-II is a level higher. It also includes advanced mesh refinement strategies, multiple phase OCP and automatic scaling. For solving the problem it uses IPOPT or SNOPT. It interfaces with MATLAB and C++, uses ADiGator [?] for automatic differentiation. This tool is commercial and closed source. Implementation in this thesis is heavily based on work of community from University of Florida.

5.2 CasADi

CasADi is an open-source tool for nonlinear optimization and algorithmic differentiation.[?][?]. interfaces with C++,Python,MATLAB. It provides a framework INE SLOVO to build the problem, and calculate the hessian and jacobian for nlp solver it does not have any mesh refinement algorithms. Commonly paired with IPOPT for nonlinear programming, but is compatible with large variety of solvers.

5.3 dymos + OpenMDAO

Python has hp mesh refinement uses OpenMDAO [21] developed at NASA Glenn research center for evaluating aircraft concepts. Lobatto and radau collocation as default [17]. According to forum post from julia discourse, which may be biased towards julia it is slower compared to infiteOpt in julia [5]. But it has mesh refinement so that is good.

5.4 Acados

[?]

5.5 Julia + JuMP/ infititeopt/ examomodels

ExaModels, JuMP

JuMP is the go to tool for algebraic modelling in julia, it is mature, versatile. Supports Linear programming, quadratic programming, mixed linear programming, Nonlinear programming. ExaModels [42] is a specialized algebraic modelling tool, it utilizes Single instruction, multiple data methods,enabling derivative evaluation on GPU accelerators. Coupled with MadNLP solver, the entire computation of nonlinear optimisation can be run on a GPU.

Julia options

1. Julia + jump

2. Julia + jump + infinte opt

3. julia + ExaModels

Infinite opt already has various collocation methods implemented. For learning purposes and in fear that in future down the road i would realise that infiniteopt blocks all the stuff i want to do i decided to use jump, without infinite opt.

No julia package has mesh refinement I chose julia + jump,

Table 5.1: Comparison of Julia (JuMP + InfiniteOpt) and Python (OpenMDAO + dymos) stacks for optimal control problem formulation.

Aspect	Julia + JuMP + InfiniteOpt	Python + OpenMDAO + dymos
Language / runtime	Julia, JIT-compiled, no GIL	Python, interpreted, GIL (NumPy releases)
License	MIT (all components)	Apache 2.0
Algebraic modelling style	Macro-based (<code>@variable</code> , <code>@constraint</code> , <code>@objective</code>)	Component graph (<code>ExplicitComponent</code> , <code>Group</code> , promotes/connects)
Derivatives	AD via <code>MathOptInterface</code> ; sparsity from expression graph	MAUD unified derivatives; mix analytic / AD / FD / complex-step per component, framework chains them
Transcription methods	Finite difference (Forward, Central, Backward); orthogonal collocation on finite elements with Gauss-Lobatto nodes only	Radau pseudospectral, Gauss-Lobatto, explicit shooting, ExplicitShooting RK
Mesh / grid refinement	None built-in; user must resolve on refined supports	Built-in ph- and hp-adaptive grid refinement
Multi-phase trajectories	Manual; user concatenate <code>InfiniteModels</code> and link	Native <code>Trajectory</code> and <code>Phase</code> with linkage constraints
Multidisciplinary coupling	Not supported; pure infinite-dim model	Native MDO coupling with implicit/explicit components, cyclic groups, nonlinear/linear sub-solvers
NLP solvers	IPOPT, MadNLP, Knitro, SNOPT and any MOI-compatible solver	IPOPT, SNOPT, scipy via <code>py-OptSparse</code>
GPU acceleration	Yes via <code>ExaModels</code> + <code>MadNLP</code> (SIMD on GPU)	Not native
Parallelism	Native multithreading, Distributed, MPI	MPI across components
Sparse total Jacobian	Yes, from MOI expression graph	Yes, MAUD assembles sparse total Jacobian from component partials
Maturity / community	JuMP mature in OR / energy; <code>InfiniteOpt</code> newer, smaller user base	NASA-backed, large aerospace user base
Learning curve for OCP only	Low; close to mathematical formulation	Higher; OpenMDAO component model required even for single-discipline OCP

Chapter 6

My implementation, julia + JuMP

I have decided to use julia + JuMP, because I wanted to use free and open source software.

I have already done a lot in Python and julia seemed like new interesting thing to learn. Julia is supposed to be fast and high level language, bridging the gap between C++ and python. Julia packages offer a lot of ready to use mathematical tools developed by community. Julia has

JuMP [30] (Julia for Mathematical Programming) is a modelling language and collection of optimization packages in Julia. JuMP makes it possible to create decision variables and constraints symbolically and uses automatic differentiation, to supply first and second derivative to solvers. Solver used in this thesis is IPOPT [47] and MadNLP[41][40]. MadNLP also allows utilising GPU for solver computation.

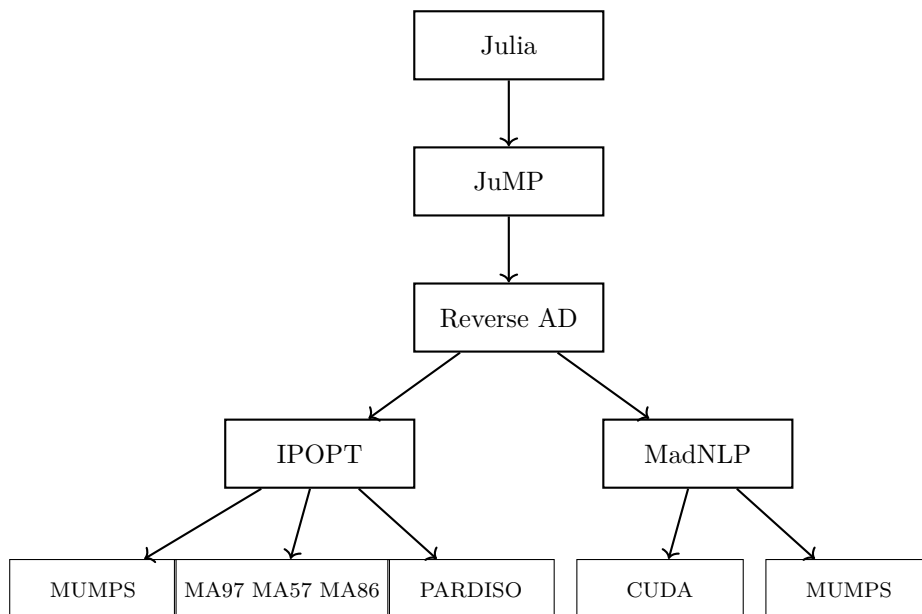


Figure 6.1: Optimization software stack

6.1 sensitivity Analysis

I have used DiffOpt.jl [?] for the sensitivity analysis. By adding vehicle parameters such as mass, lift coefficient, CoG position etc. into optimisation as decision variables fixed to a value by constraints. This adds them to the algebraic model, then the solution of minimum laptime can be differentiated by these parameters. AKO POMENOVAT TEN STROM CO JUMP VYTCORI TEN GRAF? algebraic model. Result of this is on figure 6.2

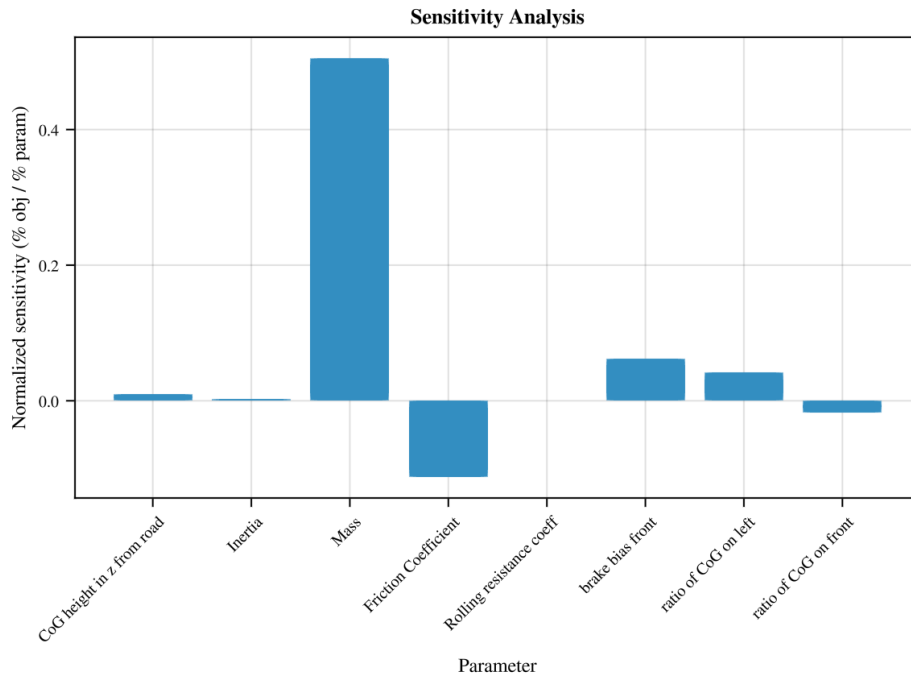


Figure 6.2: Node-interval ODE error for power-test case.



Part III

results

Chapter 7

comparison of methods

Performance of NLP solvers and frameworks is often tested on benchmark problems. like CUTEst [20] dataset?

Table 7.2: Benchmark summary statistics by transcription variant.

var <i>String</i>	runs <i>Int64</i>	ok <i>Int64</i>	fail <i>Int64</i>	rate <i>Float64</i>	mean_s <i>Float64</i>	med_s <i>Float64</i>	min_s <i>Float64</i>	max_s <i>Float64</i>
Legendre	14	13	1	92.8571	112.9541	24.1435	0.6793	987.1442
Lobatto IIIA	14	13	1	92.8571	63.1331	22.9586	0.2578	494.6361
Radau	14	14	0	100.0000	69.9256	22.0941	0.4030	544.0750

Table 7.4: Benchmark results by track, vehicle model, and transcription variant.

track <i>String</i>	car <i>String</i>	var <i>String</i>	ok <i>String</i>	status <i>String</i>	time_s <i>Float64</i>	obj <i>Float64</i>
FSCZ	singletrack	Radau	yes	solved	29.2079	50.2106
FSCZ	singletrack	Legendre	yes	solved	49.8701	50.5890
FSCZ	singletrack	Lobatto IIIA	yes	solved	29.5031	51.4710
FSCZ	twintrack	Radau	yes	solved	120.8574	57.3381
FSCZ	twintrack	Legendre	no	locally infeasible	395.5187	NaN
FSCZ	twintrack	Lobatto IIIA	no	locally infeasible	216.0335	NaN
Berlin	singletrack	Radau	yes	solved	25.5617	79.4234
Berlin	singletrack	Legendre	yes	solved	43.5952	79.4033
Berlin	singletrack	Lobatto IIIA	yes	solved	38.2088	79.1299
Berlin	twintrack	Radau	yes	solved	66.1549	77.7439
Berlin	twintrack	Legendre	yes	solved	133.1243	77.7883
Berlin	twintrack	Lobatto IIIA	yes	solved	50.8943	77.8983
Berlin	formula	Radau	yes	solved	65.8335	87.9477
Berlin	formula	Legendre	yes	solved	96.2486	88.3235
Berlin	formula	Lobatto IIIA	yes	solved	75.6741	87.9482
Berlin	bus	Radau	yes	solved	544.0750	106.9827
Berlin	bus	Legendre	yes	solved	987.1442	106.6620
Berlin	bus	Lobatto IIIA	yes	solved	494.6361	105.8631
FSG	singletrack	Radau	yes	solved	18.6264	58.2463
FSG	singletrack	Legendre	yes	solved	24.1435	58.1609
FSG	singletrack	Lobatto IIIA	yes	solved	22.9586	56.8773
FSG	twintrack	Radau	yes	solved	83.2598	69.3149
FSG	twintrack	Legendre	yes	solved	104.6524	70.3348
FSG	twintrack	Lobatto IIIA	yes	solved	91.5923	69.8878
DoubleTurn	singletrack	Radau	yes	solved	0.4030	5.7087
DoubleTurn	singletrack	Legendre	yes	solved	0.6793	5.6794
DoubleTurn	singletrack	Lobatto IIIA	yes	solved	0.2578	6.0247
DoubleTurn	twintrack	Radau	yes	solved	1.9193	6.5283
DoubleTurn	twintrack	Legendre	yes	solved	1.1882	6.5835
DoubleTurn	twintrack	Lobatto IIIA	yes	solved	2.4134	7.0205
DoubleTurn	bus	Radau	yes	solved	1.2272	8.9075
DoubleTurn	bus	Legendre	yes	solved	1.7397	8.9864
DoubleTurn	bus	Lobatto IIIA	yes	solved	1.8924	8.8676
FigureEight	singletrack	Radau	yes	solved	2.5160	14.9127
FigureEight	singletrack	Legendre	yes	solved	1.9935	14.9162
FigureEight	singletrack	Lobatto IIIA	yes	solved	1.4365	14.8296
FigureEight	twintrack	Radau	yes	solved	11.2849	16.5690
FigureEight	twintrack	Legendre	yes	solved	19.1002	16.5736
FigureEight	twintrack	Lobatto IIIA	yes	solved	8.4506	16.5319
FigureEight	bus	Radau	yes	solved	8.0315	22.8420
FigureEight	bus	Legendre	yes	solved	4.9239	22.7966
FigureEight	bus	Lobatto IIIA	yes	solved	2.8118	22.3442

Table 7.6: Linear solver benchmark summary statistics.

var <i>String</i>	runs <i>Int64</i>	ok <i>Int64</i>	fail <i>Int64</i>	rate <i>Float64</i>	mean_s <i>Float64</i>	med_s <i>Float64</i>	min_s <i>Float64</i>	max_s <i>Float64</i>
ma27	14	14	0	100.0000	94.2482	12.0524	0.3437	940.3233
ma57	14	14	0	100.0000	89.8560	11.3900	0.2677	863.0074
ma97	14	14	0	100.0000	105.2941	11.7136	0.2817	1077.0837
mumps	14	14	0	100.0000	66.2767	12.5769	0.2882	543.5226

Table 7.8: Linear solver benchmark results by track, vehicle model, and transcription variant.

track <i>String</i>	car <i>String</i>	var <i>String</i>	ok <i>String</i>	status <i>String</i>	time_s <i>Float64</i>	obj <i>Float64</i>
FSCZ	singletrack	mumps	yes	solved	10.0853	50.2106
FSCZ	singletrack	ma57	yes	solved	10.7913	50.2106
FSCZ	singletrack	ma27	yes	solved	11.0494	50.2106
FSCZ	singletrack	ma97	yes	solved	10.7454	50.2106
FSCZ	twintrack	mumps	yes	solved	102.8949	57.3381
FSCZ	twintrack	ma57	yes	solved	86.8267	57.3381
FSCZ	twintrack	ma27	yes	solved	102.6995	57.2790
FSCZ	twintrack	ma97	yes	solved	87.4077	57.3381
Berlin	singletrack	mumps	yes	solved	24.7081	79.4234
Berlin	singletrack	ma57	yes	solved	24.0319	79.4234
Berlin	singletrack	ma27	yes	solved	23.7324	79.4234
Berlin	singletrack	ma97	yes	solved	24.8796	79.4234
Berlin	twintrack	mumps	yes	solved	63.9139	77.7439
Berlin	twintrack	ma57	yes	solved	58.8653	77.7439
Berlin	twintrack	ma27	yes	solved	49.7654	77.7439
Berlin	twintrack	ma97	yes	solved	54.2876	77.7439
Berlin	formula	mumps	yes	solved	64.7728	87.9477
Berlin	formula	ma57	yes	solved	62.8802	87.9477
Berlin	formula	ma27	yes	solved	59.1794	87.9477
Berlin	formula	ma97	yes	solved	73.3883	87.9477
Berlin	bus	mumps	yes	solved	543.5226	106.9827
Berlin	bus	ma57	yes	solved	863.0074	106.9795
Berlin	bus	ma27	yes	solved	940.3233	106.9815
Berlin	bus	ma97	yes	solved	1077.0837	106.6410
FSG	singletrack	mumps	yes	solved	14.2399	58.2463
FSG	singletrack	ma57	yes	solved	11.9887	58.2463
FSG	singletrack	ma27	yes	solved	13.0554	58.2463
FSG	singletrack	ma97	yes	solved	12.6819	58.2463
FSG	twintrack	mumps	yes	solved	80.0414	69.3149
FSG	twintrack	ma57	yes	solved	117.0157	69.3471
FSG	twintrack	ma27	yes	solved	98.0660	69.3140
FSG	twintrack	ma97	yes	solved	111.5260	69.3140
DoubleTurn	singletrack	mumps	yes	solved	0.2882	5.7087
DoubleTurn	singletrack	ma57	yes	solved	0.2677	5.7087
DoubleTurn	singletrack	ma27	yes	solved	0.3437	5.7087
DoubleTurn	singletrack	ma97	yes	solved	0.2817	5.7087
DoubleTurn	twintrack	mumps	yes	solved	2.0100	6.5283
DoubleTurn	twintrack	ma57	yes	solved	2.6468	6.5283
DoubleTurn	twintrack	ma27	yes	solved	1.9489	6.5283
DoubleTurn	twintrack	ma97	yes	solved	2.5591	6.5283
DoubleTurn	bus	mumps	yes	solved	1.4624	8.9075
DoubleTurn	bus	ma57	yes	solved	1.3533	8.9075
DoubleTurn	bus	ma27	yes	solved	1.2714	8.9075
DoubleTurn	bus	ma97	yes	solved	1.3283	8.9075
FigureEight	singletrack	mumps	yes	solved	0.8930	14.9127
FigureEight	singletrack	ma57	yes	solved	0.9078	14.9127
FigureEight	singletrack	ma27	yes	solved	0.8553	14.9127
FigureEight	singletrack	ma97	yes	solved	0.8441	14.9127
FigureEight	twintrack	mumps	yes	solved	10.9140	16.5690
FigureEight	twintrack	ma57	yes	solved	9.9301	16.5690
FigureEight	twintrack	ma27	yes	solved	9.2067	16.5690
FigureEight	twintrack	ma97	yes	solved	9.3497	16.5690
FigureEight	bus	mumps	yes	solved	8.1270	22.8420
FigureEight	bus	ma57	yes	solved	7.4718	22.8420
FigureEight	bus	ma27	yes	solved	7.9787	22.8420
FigureEight	bus	ma97	yes	solved	7.7546	22.8420

Table 7.10: NLP solver / backend benchmark summary statistics (IPOPT-MUMPS, MadNLP-CPU, MadNLP-GPU).

var <i>String</i>	runs <i>Int64</i>	ok <i>Int64</i>	fail <i>Int64</i>	rate <i>Float64</i>	mean_s <i>Float64</i>	med_s <i>Float64</i>	min_s <i>Float64</i>	max_s <i>Float64</i>
ipopt-mumps	14	14	0	100.0000	123.8362	12.2045	0.2810	1322.9633
madnlp-cpu	14	14	0	100.0000	124.3906	16.8290	0.2802	1351.5053
madnlp-gpu	14	14	0	100.0000	69.4331	18.0661	0.4816	566.9196

Table 7.12: NLP solver / backend benchmark results by track, vehicle model, and transcription variant.

track <i>String</i>	car <i>String</i>	var <i>String</i>	ok <i>String</i>	status <i>String</i>	time_s <i>Float64</i>	obj <i>Float64</i>
FSCZ	singletrack	ipopt-mumps	yes	solved	10.3733	50.2106
FSCZ	singletrack	madnlp-cpu	yes	solved	18.5945	50.2106
FSCZ	singletrack	madnlp-gpu	yes	solved	72.7663	50.2098
FSCZ	twintrack	ipopt-mumps	yes	solved	106.7641	57.3381
FSCZ	twintrack	madnlp-cpu	yes	solved	100.3124	57.2490
FSCZ	twintrack	madnlp-gpu	yes	solved	73.8696	57.2521
Berlin	singletrack	ipopt-mumps	yes	solved	24.2003	79.4234
Berlin	singletrack	madnlp-cpu	yes	solved	18.5749	79.4234
Berlin	singletrack	madnlp-gpu	yes	solved	22.4556	79.4214
Berlin	twintrack	ipopt-mumps	yes	solved	85.2394	77.7417
Berlin	twintrack	madnlp-cpu	yes	solved	58.1692	77.7442
Berlin	twintrack	madnlp-gpu	yes	solved	42.9623	77.7421
Berlin	formula	ipopt-mumps	yes	solved	63.1405	87.9477
Berlin	formula	madnlp-cpu	yes	solved	60.3523	87.9477
Berlin	formula	madnlp-gpu	yes	almost locally solved	80.3682	87.9191
Berlin	bus	ipopt-mumps	yes	solved	1322.9633	107.2120
Berlin	bus	madnlp-cpu	yes	solved	1351.5053	106.6728
Berlin	bus	madnlp-gpu	yes	solved	566.9196	106.9743
FSG	singletrack	ipopt-mumps	yes	solved	12.3827	58.2463
FSG	singletrack	madnlp-cpu	yes	solved	13.5735	58.2463
FSG	singletrack	madnlp-gpu	yes	solved	13.6767	58.2455
FSG	twintrack	ipopt-mumps	yes	solved	83.7747	69.3149
FSG	twintrack	madnlp-cpu	yes	solved	89.3044	69.4039
FSG	twintrack	madnlp-gpu	yes	solved	78.0973	69.3745
DoubleTurn	singletrack	ipopt-mumps	yes	solved	0.2810	5.7087
DoubleTurn	singletrack	madnlp-cpu	yes	solved	0.2802	5.7087
DoubleTurn	singletrack	madnlp-gpu	yes	solved	0.4816	5.7086
DoubleTurn	twintrack	ipopt-mumps	yes	solved	2.0090	6.5283
DoubleTurn	twintrack	madnlp-cpu	yes	solved	3.8171	6.5463
DoubleTurn	twintrack	madnlp-gpu	yes	solved	2.4273	6.5282
DoubleTurn	bus	ipopt-mumps	yes	solved	1.2878	8.9075
DoubleTurn	bus	madnlp-cpu	yes	solved	1.2564	8.9075
DoubleTurn	bus	madnlp-gpu	yes	solved	1.5964	8.9001
FigureEight	singletrack	ipopt-mumps	yes	solved	0.8911	14.9127
FigureEight	singletrack	madnlp-cpu	yes	solved	0.9631	14.9127
FigureEight	singletrack	madnlp-gpu	yes	solved	1.1818	14.9124
FigureEight	twintrack	ipopt-mumps	yes	solved	12.0263	16.5690
FigureEight	twintrack	madnlp-cpu	yes	solved	15.0831	16.5658
FigureEight	twintrack	madnlp-gpu	yes	solved	6.4857	16.5652
FigureEight	bus	ipopt-mumps	yes	solved	8.3730	22.8420
FigureEight	bus	madnlp-cpu	yes	solved	9.6822	22.7936
FigureEight	bus	madnlp-gpu	yes	solved	8.7756	22.7984

Chapter 8

Conclusion

All goals were achieved. One thing which was not a goal but, would be super useful if i managed to do a validation on real world data. Validation of sensitivity analysis. I wanted to do it for a long time but there was not enough of it.

Julia was interesting language to learn, it offers blazing fast execution, but that comes with caveats. Debugging is way more complicated. When working on a smaller project, the debugging is not such a pain, but when the project is bigger debugging is not user friendly. Debugging in python and MATLAB is way easier. Julias selling point is that it is just in time compiled language, rivalling C. That is true BUT you have to know what you are doing, if you dont declare types in functions it is as slow as python.

At the begging of this project I wanted a more complex model, I wanted to go deeper with complexity than my bachelor thesis. That is add pitch, roll and full model of suspension into the simulation. Also this simulation does not account for gearboxes with discrete stages, and no combustion engine.

Believing 100% this tool and designing car according to it would most likely lead to undriveable vehicle, as there is no force pushing for stability driveability in NLP [36] deals also with this. The optimal driving created by NLP is unachievable by real driver, as the vehicle is on its limit and one mistake woul overload tires and cause failure.

Track model is also heavily simplified, neglecting slope and bank of the track.



Appendices

Appendix A

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